Describing Motion in a Straight Line

Comprehensive Notes: From Foundational Concepts to Advanced Olympiad Mechanics

Source Materials: NCERT Physics Chapter 4 (iesc104.pdf) & Allen Physics Module (Motion.pdf)

Executive Overview & Learning Roadmap

This study guide synthesizes the fundamental principles of 1D kinematics from foundational physical definitions to advanced multi-stage problem-solving techniques.

- **Foundations:** Frames of reference, origins, coordinate systems, scalars vs. vectors, path length vs. displacement.
- **Kinematic Rates:** Instantaneous vs. average speed, velocity vectors, average and instantaneous acceleration.
- **Graphical Dynamics:** Interpretation of slopes ($\tan\theta$) and area integrals across $x-t$, $v-t$, and $a-t$ curves.
- **Analytical Kinematics:** Rigorous algebraic derivations of constant-acceleration equations, n^{th} second displacement, and midpoint velocity theorems.
- **Gravitational Motion:** Vertical free fall, upward projection, symmetry principles, tower/rising balloon mechanics.
- **Olympiad Workshop:** Worked numerical challenges, edge-case analysis, and comprehensive practice sets.

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1 Module 1: Reference Frames, Coordinates, and Spatial Position

1.1 Mechanics: Kinematics vs. Dynamics

The study of physical motion in classical mechanics is traditionally divided into two core branches:

1. Kinematics: The branch of mechanics concerned with describing the motion of points, bodies, or systems of bodies without considering the forces that cause the motion. It focuses on position, displacement, time, velocity, and acceleration.
2. Dynamics: The branch of mechanics that analyzes the causes of motion, examining how forces and torques influence the state of movement.

1.2 Concept of Rest, Motion, and Relativity of Observer

Motion is defined as a continuous change in the position of an object relative to its surroundings over time. A fundamental tenet of kinematics is that motion is relative, not absolute. An object is in motion with respect to one frame of reference while simultaneously at rest with respect to another.

- Example: A passenger seated inside a moving train is at rest relative to fellow passengers, but in rapid motion relative to an observer standing on the railway platform.

1.3 Establishing a Frame of Reference & Coordinate Systems

To quantitatively describe the location of a physical particle, a frame of reference must be specified. A 1D reference frame consists of:

1. A Reference Point (Origin, O): A fixed, arbitrary landmark chosen to represent zero position ($x = 0$).
2. A Direction Axis: A straight spatial line passing through O.
3. A Sign Convention: Positions to the right (or forward) of O are assigned positive values ($+x$), while positions to the left (or backward) are assigned negative values ($-x$).
4. A Time Scale: A clock measuring elapsed time t from an initial instant ($t = 0$).

Conceptual Illustration: Reference Frame Specification

Suppose a car is located 100 m to the right of a fixed tree chosen as the origin O. Its position is represented as $x = +100$ m. If the car moves 75 m to the left of the tree, its position becomes $x = -75$ m. Position is a variable state quantity defined strictly relative to O.

2 Module 2: Scalar & Vector Quantities – Distance vs. Displacement

2.1 Physical Quantities: Scalars and Vectors

Physical quantities in kinematics are categorized based on their directional dependence:

- Scalar Quantities: Physical variables fully specified by a numerical magnitude and an appropriate unit. Scalars have no direction.

  Examples: Mass (m), Time (t), Distance (s), Speed (v), Energy (E), Temperature (T). (1)

- Vector Quantities: Physical variables requiring both a numerical magnitude and a spatial direction for complete description.

  Examples: Position $\vec{x}$, Displacement ($\Delta\vec{x}$), Velocity $\vec{v}$, Acceleration $\vec{a}$, Force ($\vec{F}$). (2)

2.2 Distance (Path Length)

Distance is the total length of the actual spatial path traversed by a moving object between its initial and final positions during a given time interval.

- Nature: Scalar quantity.
- Path Dependence: Strictly depends on the specific path chosen.
- Mathematical Constraint: Always positive ($s > 0$) for a body in motion. It can never decrease with time; $\frac{ds}{dt} \geq 0$.
- SI Unit: Metre (m).

2.3 Displacement

Displacement is defined as the net change in position of a particle during a specified time interval. It represents the shortest directed straight-line segment from the initial position to the final position.

$$\Delta x = x_f - x_i \tag{3}$$

where x_f is the final position coordinate and x_i is the initial position coordinate.

- Nature: Vector quantity. Its direction points from initial position x_i toward final position x_f.
- Path Independence: Independent of the actual path traversed; dependent solely on initial and final coordinates.
- Sign Properties: Can be positive, zero, or negative depending on the direction of net movement relative to origin O.
- Zero Displacement: If a particle returns to its starting point ($x_f = x_i$), its displacement is identically zero ($\Delta x = 0$), even though total distance travelled $s > 0$.

2.4 Fundamental Inequality: Distance vs. Displacement

The total distance travelled is always greater than or equal to the magnitude of displacement:

$$\text{Distance } (s) \geq |\text{Displacement } (\Delta x)| \tag{4}$$

Conditions for Equality

The magnitude of displacement is exactly equal to the total distance travelled if and only if:

1. The motion occurs strictly along a straight-line path, AND
2. The object moves in a single direction without turning back or reversing its motion.

Whenever a particle reverses direction or follows a curved path, Distance > |Displacement|.

2.5 Detailed Case Studies & Geometric Examples

2.5.1 Case Study 1: The Track Athlete (NCERT Example)

An athlete starts running from origin O ($t = 0$ s), reaches point B ($x = 40$ m at $t = 4$ s), then reaches point A ($x = 100$ m at $t = 10$ s), and finally turns back along the same track to stop at point B ($x = 40$ m at $t = 16$ s).

- Total Distance Travelled:

$$s = OA + AB = 100\text{ m} + (100\text{ m} - 40\text{ m}) = 100\text{ m} + 60\text{ m} = 160\text{ m} \tag{5}$$

- **Net Displacement:**

$$\Delta x = x_f - x_i = x_B - x_O = 40\ \text{m} - 0\ \text{m} = +40\ \text{m} \tag{6}$$

Notice that 160 m > 40 m.

2.5.2 Case Study 2: Two-Dimensional Corner Movement

A person walks 30 m due East (AB) and then 40 m due North (BC).

- **Total Distance:** $s = AB + BC = 30\ \text{m} + 40\ \text{m} = 70\ \text{m}$.
- **Magnitude of Displacement:** Using the Pythagorean theorem:

$$|\vec{\Delta x}| = AC = \sqrt{AB^2 + BC^2} = \sqrt{30^2 + 40^2} = \sqrt{900 + 1600} = \sqrt{2500} = 50\ \text{m} \tag{7}$$

3 Module 3: Kinematic Rates — Speed, Velocity, and Acceleration

3.1 Uniform vs. Non-Uniform Motion

- **Uniform Motion:** An object covers equal distances (or undergoes equal displacements) in equal intervals of time, no matter how small these time intervals may be. Velocity is constant, and acceleration is zero ($a = 0$).
- **Non-Uniform Motion:** An object covers unequal distances in equal intervals of time, or changes its direction of movement. Velocity varies with time, indicating non-zero acceleration ($a \neq 0$).

3.2 Speed: Average and Instantaneous

Speed is the rate at which distance is covered with respect to time. It is a non-negative scalar quantity.

3.2.1 Average Speed

Average speed (v_{avg}) is the ratio of the total distance travelled to the total time elapsed:

$$v_{avg} = \frac{\text{Total Distance Travelled}}{\text{Total Time Elapsed}} = \frac{s_{total}}{\Delta t} \tag{8}$$

3.2.2 Instantaneous Speed

Instantaneous speed is the speed of an object at a specific instant of time t. Mathematically, it is the limiting value of average speed as time interval $\Delta t \to 0$:

$$v_{inst} = \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt} \tag{9}$$

Note: The speedometer in automobiles measures instantaneous speed. In uniform motion, instantaneous speed is identically equal to average speed at all instants.

3.3 Derivations for Special Cases of Average Speed

3.3.1 Case A: Equal Distance Segments (Harmonic Mean Derivation)

Consider an object covering two equal distances ($s_1 = s_2 = s$) with distinct constant speeds v_1 and v_2.

$$\text{Time taken for first half: } t_1 = \frac{s}{v_1}, \quad \text{Time for second half: } t_2 = \frac{s}{v_2} \tag{10}$$

$$v_{avg} = \frac{s + s}{t_1 + t_2} = \frac{2s}{\frac{s}{v_1} + \frac{s}{v_2}} = \frac{2s}{s\left(\frac{v_1 + v_2}{v_1 v_2}\right)} = \frac{2v_1 v_2}{v_1 + v_2} \tag{11}$$

Thus, when equal distance segments are covered, average speed is the harmonic mean of individual speeds.

For three equal distance segments ($s_1 = s_2 = s_3 = s$) covered at speeds v_1, v_2, v_3:

$$v_{avg} = \frac{3v_1v_2v_3}{v_1v_2 + v_2v_3 + v_3v_1} \tag{12}$$

3.3.2 Case B: Equal Time Intervals (Arithmetic Mean Derivation)

Consider an object travelling with speeds $v_1, v_2, \ldots, v_n$ during equal time intervals $t_1 = t_2 = \cdots = t_n = t$.

$$\text{Distance covered: } s_1 = v_1t, \quad s_2 = v_2t, \quad \ldots \quad s_n = v_nt \tag{13}$$

$$v_{avg} = \frac{s_1 + s_2 + \cdots + s_n}{nt} = \frac{(v_1 + v_2 + \cdots + v_n)t}{nt} = \frac{v_1 + v_2 + \cdots + v_n}{n} \tag{14}$$

When time intervals are equal, average speed is the arithmetic mean of individual speeds.

3.4 Velocity: Average and Instantaneous

Velocity is the rate of change of position with respect to time. It is a vector quantity having both magnitude and direction.

3.4.1 Average Velocity

Average velocity ($\vec{v}_{avg}$) is the net displacement divided by total time:

$$\vec{v}_{avg} = \frac{\Delta\vec{x}}{\Delta t} = \frac{x_f - x_i}{t_f - t_i} \tag{15}$$

3.4.2 Instantaneous Velocity

Instantaneous velocity ($\vec{v}$) is the time derivative of position vector $\vec{x}$:

$$\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta\vec{x}}{\Delta t} = \frac{d\vec{x}}{dt} \tag{16}$$

The magnitude of instantaneous velocity is always equal to the instantaneous speed.

3.5 Acceleration

Acceleration ($\vec{a}$) is the rate of change of velocity with respect to time. It is a vector quantity with SI unit m/s^2 ($m \cdot s^{-2}$).

3.5.1 Average Acceleration

$$\vec{a}_{avg} = \frac{\Delta\vec{v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_i}{t_2 - t_1} = \frac{v - u}{t} \tag{17}$$

where u is initial velocity and v is final velocity over time interval t.

3.5.2 Instantaneous Acceleration

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{x}}{dt^2} = v\frac{dv}{dx} \tag{18}$$

Direction of Acceleration vs. Motion

- **Speeding Up (Acceleration):** When velocity $\vec{v}$ and acceleration $\vec{a}$ have the same sign (same direction), the speed of the body increases.
- **Slowing Down (Retardation / Deceleration):** When velocity $\vec{v}$ and acceleration $\vec{a}$

have opposite signs (opposite directions), the speed of the body decreases.

4 Module 4: Graphical Analysis of Motion

4.1 Geometric Principles: Slopes and Integrals

Graphical visual representation provides powerful qualitative and quantitative insight into kinematics.

1. Slope ($\tan\theta$): The slope of a tangent line to a curve on an $x-y$ graph represents the derivative $\frac{dy}{dx}$.

$$\text{Slope} = \tan\theta = \frac{\text{Perpendicular}}{\text{Base}} = \frac{y_2 - y_1}{x_2 - x_1} \tag{19}$$

2. Area Under Curve: The integral $\int_R y\,dx$ represents physical quantities formed by multiplying the ordinate and abscissa units.

4.2 Position-Time ($x-t$) & Distance-Time ($s-t$) Graphs

The slope of a position-time graph yields the instantaneous velocity ($\vec{v} = \frac{dx}{dt}$).

Table 1: Interpretation of Position-Time ($x-t$) Graphs

Graph Profile	Mathematical Form	Physical State of Motion
Horizontal line parallel to t-axis	$x(t) = \text{constant}$	Object at rest ($v = 0$).
Straight line with constant slope	$x(t) = mt + c$	Uniform velocity ($a = 0$).
Curve bending upward (θ increasing)	$\frac{d^2x}{dt^2} > 0$	Uniformly accelerated motion ($a > 0$).
Curve bending downward (θ decreasing)	$\frac{d^2x}{dt^2} < 0$	Retarded motion ($a < 0$).
Slope equal to zero at peak/valley	$\frac{dx}{dt} = 0$	Turning point (instantaneous rest).

Crucial Rule for Distance-Time Graphs

A distance-time ($s-t$) graph can NEVER decrease with time because distance is monotonic non-decreasing ($s \geq 0$, $\frac{ds}{dt} \geq 0$). A graph showing a decreasing distance curve is physically impossible!

4.3 Velocity-Time ($v-t$) Graphs

1. Slope of $v-t$ Graph: Represents acceleration ($a = \frac{dv}{dt}$).
2. Area Under $v-t$ Graph:
   - Signed Area Sum: Represents Net Displacement ($\Delta x = \int_R v\,dt$).
   - Absolute Area Sum: Represents Total Distance ($s = \int_R |v|\,dt$).

Displacement vs. Distance from $v-t$ Area

Consider a $v-t$ graph consisting of positive area region A_1 above the t-axis, negative area region A_2 below the t-axis, and positive area region A_3 above the t-axis:

$$\text{Displacement } (\Delta x) = A_1 - A_2 + A_3, \qquad \text{Distance } (s) = A_1 + A_2 + A_3 \tag{20}$$

4.4 Acceleration-Time ($a-t$) Graphs

The area under an acceleration-time graph represents the net change in velocity:

$$\text{Area under } a-t \text{ curve} = \int_{t_1}^{t_2} a\,dt = v_f - v_i = \Delta v \tag{21}$$

5 Module 5: Derivations of Kinematic Equations

5.1 Conditions for Validity

The classical kinematic equations derived below are valid STRICTLY for motion with constant (uniform) acceleration along a straight line.

5.2 Derivation of Primary Kinematic Equations

5.2.1 First Equation of Motion ($v = u + at$)

By definition of constant acceleration:

$$a = \frac{v-u}{t-0} \Rightarrow at = v - u \Rightarrow v = u + at \tag{22}$$

5.2.2 Second Equation of Motion ($s = ut + \frac{1}{2}at^2$)

For uniform acceleration, average velocity is the arithmetic average of initial and final velocities:

$$v_{avg} = \frac{u+v}{2} \tag{23}$$

Displacement s is average velocity multiplied by time t:

$$s = v_{avg} \cdot t = \left(\frac{u+v}{2}\right)t \tag{24}$$

Substituting $v = u + at$ into the displacement relation:

$$s = \left(\frac{u+(u+at)}{2}\right)t = \left(\frac{2u+at}{2}\right)t = \left(u + \frac{1}{2}at\right)t = ut + \frac{1}{2}at^2 \tag{25}$$

5.2.3 Third Equation of Motion ($v^2 = u^2 + 2as$)

From $v = u + at$, time elapsed is $t = \frac{v-u}{a}$. Substitute this into $s = \left(\frac{u+v}{2}\right)t$:

$$s = \left(\frac{v+u}{2}\right)\left(\frac{v-u}{a}\right) = \frac{v^2-u^2}{2a} \tag{26}$$

$$2as = v^2 - u^2 \Rightarrow v^2 = u^2 + 2as \tag{27}$$

5.3 Displacement in the nth Specific Second

To find displacement $s_{n^{th}}$ occurring precisely during the interval between $t = (n-1)$ and $t = n$ seconds:

$$s_{n^{th}} = s_n - s_{n-1} \tag{28}$$

where $s_n = un + \frac{1}{2}an^2$ and $s_{n-1} = u(n-1) + \frac{1}{2}a(n-1)^2$.

Algebraic Expansion:

$$s_{n^{th}} = \left[un + \frac{1}{2}an^2\right] - \left[u(n-1) + \frac{1}{2}a(n-1)^2\right] \tag{29}$$

$$s_{n^{th}} = un - u(n-1) + \frac{1}{2}a\left[n^2 - (n^2 - 2n + 1)\right] \tag{30}$$

$$s_{n^{th}} = u(n - n + 1) + \frac{1}{2}a(2n-1) = u + \frac{a}{2}(2n-1) \tag{31}$$

5.4 Midpoint Velocity Theorem Derivation

Consider a particle moving with constant acceleration a along line segment AB. Initial velocity at A is u, final velocity at B is v, and C is the midpoint of AB ($AC = CB = s/2$).

$$\text{For segment } AC: \quad v_C^2 = u^2 + 2a\left(\frac{s}{2}\right) = u^2 + as \Rightarrow as = v_C^2 - u^2 \tag{32}$$

$$\text{For segment } CB: \quad v^2 = v_C^2 + 2a\left(\frac{s}{2}\right) = v_C^2 + as \Rightarrow as = v^2 - v_C^2 \tag{33}$$

Equating the two expressions for as:

$$v_C^2 - u^2 = v^2 - v_C^2 \Rightarrow 2v_C^2 = u^2 + v^2 \Rightarrow v_C = \sqrt{\frac{u^2 + v^2}{2}} \tag{34}$$

6 Module 6: Free Fall & Vertical Motion Under Gravity

6.1 Historical Foundations: Aristotle vs. Galileo

Until 1600 AD, Aristotelean physics posited that heavy objects fall faster than light ones. Italian physicist Galileo Galilei disproved this via experiments, establishing that in the absence of air resistance, all bodies experience the exact same constant acceleration toward Earth, regardless of mass.

6.2 Standard Values and Sign Conventions

Acceleration due to gravity (g) acts vertically downward toward the Earth's center:

$$g \approx 9.8\ \text{m/s}^2 \approx 980\ \text{cm/s}^2 \approx 32\ \text{ft/s}^2 \quad (\text{For problem solving, often } g \approx 10\ \text{m/s}^2) \tag{35}$$

Sign Convention: Taking upward direction as $+y$ and downward as $-y$:

$$\text{Acceleration vector: } a = -g \quad (\text{Always directed downward}) \tag{36}$$

6.3 Formulas for Vertical Motion Scenarios

Table 2: Kinematic Equations for Free-Fall and Vertical Projection

Motion Case	Initial Conditions	Governing Equations	Key Formulas
Dropped from Rest	$u = 0, a = +g$ (downward)	$v = gt,\ y = \frac{1}{2}gt^2,\ v^2 = 2gy$	$t_{impact} = \sqrt{\frac{2H}{g}},\ v_{impact} = \sqrt{2gH}$
Thrown Downward	$u = u_0, a = +g$	$v = u + gt,\ y = ut + \frac{1}{2}gt^2$	$v_{impact} = \sqrt{u^2 + 2gH}$
Thrown Upward	$u = +u_0, a = -g$	$v = u - gt,\ y = ut - \frac{1}{2}gt^2$	$H_{max} = \frac{u^2}{2g},\ T_{flight} = \frac{2u}{g}$

6.4 Symmetry Principles of Vertical Motion

For an object thrown vertically upward returning to the same projection level:

1. Time Symmetry: Time of ascent equals time of descent ($t_a = t_d = \frac{u}{g}$).
2. Speed Symmetry: Speed of impact at projection level equals initial projection speed ($|v_{return}| = |u_0|$).
3. Velocity Vector Symmetry: $\vec{v}_{return} = -\vec{u}_0$.

6.5 Special Case: Projectile Thrown Upward from Tower / Balloon

When a stone is projected upward with initial velocity $+u$ from a tower of height H (or dropped from a balloon ascending at rate $+u$ at height H), taking the projection point as origin:

$$\text{Net displacement at ground: } y = -H, \quad \text{Acceleration: } a = -g \tag{37}$$

$$\text{Displacement Equation: } -H = ut - \frac{1}{2}gt^2 \implies \frac{1}{2}gt^2 - ut - H = 0 \tag{38}$$

Solving this quadratic yields the total flight time t. Velocity upon striking the ground is:

$$v_{impact} = \sqrt{u^2 + 2gH} \tag{39}$$

7 Module 7: Advanced Olympiad Mechanics & Calculus Kinematics

7.1 Relative Kinematics in 1D

When two particles A and B move along the same straight line with positions x_A, x_B, velocities v_A, v_B, and accelerations a_A, a_B:

$$\text{Relative Position: } x_{AB} = x_A - x_B \tag{40}$$

$$\text{Relative Velocity: } v_{AB} = v_A - v_B \tag{41}$$

$$\text{Relative Acceleration: } a_{AB} = a_A - a_B \tag{42}$$

If $a_{AB} = 0$ (both bodies experience identical acceleration, such as gravity g), the relative velocity v_{AB} remains constant throughout the motion!

7.2 Calculus Formalism for Non-Uniform Acceleration

When acceleration varies with position or time ($a = a(t)$ or $a = a(x)$):

$$x(t) = x_0 + \int_0^t v(t')\,dt', \qquad v(t) = u + \int_0^t a(t')\,dt' \tag{43}$$

$$a(x) = v\frac{dv}{dx} \implies \int_{x_0}^{x} a(x')\,dx' = \int_u^v v'\,dv' = \frac{v^2 - u^2}{2} \tag{44}$$

7.3 Sound Propagation in Gravitational Well (Splash Problems)

When a stone is dropped into a well of depth d and the sound of the splash is heard after total time T:

1. Time for stone to fall to water surface: $t_1 = \sqrt{\frac{2d}{g}}$.
2. Time for sound wave to travel up well depth d at constant speed v_s: $t_2 = \frac{d}{v_s}$.

$$\text{Total Elapsed Time: } T = t_1 + t_2 = \sqrt{\frac{2d}{g}} + \frac{d}{v_s} \tag{45}$$

8 Module 8: Comprehensive Olympiad Problem Workshop

Problem 1: Half-Journey Train Speed Challenge (Module Source)

Problem Statement: On a 60 km track, a train travels the first 30 km at a uniform speed of 30 km/h. How fast must the train travel the next 30 km to average 40 km/h for the entire trip?

Solution: This is a half-journey equal distance case ($s_1 = s_2 = 30$ km).

$$v_{avg} = \frac{2v_1v_2}{v_1 + v_2} \Rightarrow 40 = \frac{2(30)v_2}{30 + v_2} \tag{46}$$

$$40(30 + v_2) = 60v_2 \Rightarrow 1200 + 40v_2 = 60v_2 \Rightarrow 20v_2 = 1200 \Rightarrow v_2 = 60 \text{ km/h} \tag{47}$$

Answer: The train must travel the second half at 60 km/h.

Problem 2: Cricket Ball Pitch Transit Time Analysis

Problem Statement: A bowler delivers 6 balls in an over with recorded speeds of 44,42,46,44,36,38 m/s. If the pitch is 20 m long, calculate the average time taken for the ball to reach the batsman.

Solution: Equal time intervals case for deliveries. Calculate mean bowling speed:

$$v_{avg} = \frac{44 + 42 + 46 + 44 + 36 + 38}{6} = \frac{250}{6} \approx 41.67 \text{ m/s} \tag{48}$$

Time taken to cover pitch distance $s = 20$ m:

$$t = \frac{s}{v_{avg}} = \frac{20}{41.67} \approx 0.48 \text{ s} \tag{49}$$

Answer: Average pitch transit time is 0.48 seconds.

Problem 3: Acceleration from n^{th} Second Distance

Problem Statement: A particle starts from rest ($u = 0$) and travels 150 cm (1.5 m) in the 5th second of its journey. Find its acceleration.

Solution: Use the nth second formula with $n = 5$, $u = 0$, $s_{5th} = 1.5$ m:

$$s_{n^{th}} = u + \frac{a}{2}(2n - 1) \Rightarrow 1.5 = 0 + \frac{a}{2}(2(5) - 1) = \frac{9a}{2} \tag{50}$$

$$9a = 3.0 \Rightarrow a = \frac{3}{9} = \frac{1}{3} \approx 0.33 \text{ m/s}^2 \tag{51}$$

Answer: Acceleration $a = 0.33$ m/s^2 (33.3 cm/s^2).

Problem 4: Multi-Interval Kinematic Reconstruction

Problem Statement: A body covers 200 cm in the first 2 seconds and 220 cm in the next 4 seconds under constant acceleration. Find its velocity at the end of the 7th second.

Solution: Let initial velocity be u (cm/s) and acceleration be a (cm/s^2).

1. First interval ($t_1 = 2$ s, $s_1 = 200$ cm):

$$200 = u(2) + \frac{1}{2}a(2)^2 = 2u + 2a \Rightarrow u + a = 100 \quad -(\text{Eq. } 1) \tag{52}$$

2. Cumulative interval ($t_2 = 2 + 4 = 6$ s, $s_2 = 200 + 220 = 420$ cm):

$$420 = u(6) + \frac{1}{2}a(6)^2 = 6u + 18a \Rightarrow u + 3a = 70 \quad -(\text{Eq. } 2) \tag{53}$$

Subtract Eq. 1 from Eq. 2:

$$(u + 3a) - (u + a) = 70 - 100 \Rightarrow 2a = -30 \Rightarrow a = -15 \text{ cm/s}^2 \tag{54}$$

Substitute $a = -15$ into Eq. 1:

$$u - 15 = 100 \Rightarrow u = 115 \text{ cm/s} \tag{55}$$

Velocity at $t = 7$ s:

$$v_7 = u + at = 115 + (-15)(7) = 115 - 105 = 10 \text{ cm/s} \tag{56}$$

Answer: Velocity at the end of 7th second is 10 cm/s.

Problem 5: Tower Height from Double Crossing Interval

Problem Statement: A ball is projected vertically upward from the foot of a tower. It crosses the top of the tower twice at an interval of 6 s and reaches the ground after a total time of 12 s. Calculate the height of the tower ($g = 10$ m/s^2).
Solution: Total flight time $T = 12$ s $= \frac{2u}{g} \Rightarrow 2u = 12(10) = 120 \Rightarrow u = 60$ m/s. The time gap of 6 s between crossing the tower top means 3 s ascending above the top and 3 s descending back to the top. Thus, time taken to reach tower top $t_1 = 6 - 3 = 3$ s. Using displacement equation $h = ut_1 - \frac{1}{2}gt_1^2$:

$$h = (60)(3) - \frac{1}{2}(10)(3)^2 = 180 - 45 = 135 \text{ m} \tag{57}$$

Answer: Height of the tower is 135 metres.

Problem 6: Sequential Vertical Projections (Meeting Point)

Problem Statement: Body A is projected upward with velocity 98 m/s. Body B is projected upward from the same point with the same velocity after a delay of 4 s. Find the time after which they meet ($g = 9.8$ m/s^2).
Solution: Let flight time of body A be t seconds. Then flight time of body B is $(t - 4)$ seconds. When they meet, their displacements are equal ($s_A = s_B$):

$$ut - \frac{1}{2}gt^2 = u(t-4) - \frac{1}{2}g(t-4)^2 \tag{58}$$

Substitute $u = 98$ m/s and $g = 9.8$ m/s^2:

$$98t - 4.9t^2 = 98t - 392 - 4.9(t^2 - 8t + 16) \tag{59}$$

$$-4.9t^2 = -392 - 4.9t^2 + 39.2t - 78.4 \Rightarrow 39.2t = 470.4 \Rightarrow t = 12 \text{ seconds} \tag{60}$$

Answer: The bodies meet 12 seconds after the launch of the first body.

9 Module 9: Formula Master Reference & Practice Questions

9.1 Formula Summary Table

9.2 Olympiad Practice Problem Set

1. Conceptual Challenge: Can a particle have zero velocity and non-zero acceleration simultaneously? Give an example. (Hint: Topmost point of vertical throw).

2. Conceptual Challenge: Is it possible for a body to have varying velocity while maintaining constant speed? (Hint: Uniform circular motion or reversing direction).

3. Numerical Problem: A stone dropped from a cliff travels 45 m in the last second before hitting the water surface. Calculate the total height of the cliff ($g = 10$ m/s^2).

Table 3: Master Formula Table for Straight Line Kinematics

Physical Quantity / Relation	Mathematical Formula
Displacement	$\Delta x = x_f - x_i$
Average Speed	$v_{avg} = \frac{\text{Total Distance}}{\text{Total Time}}$
Average Speed (Half Distance)	$v_{avg} = \frac{2v_1v_2}{v_1+v_2}$
Average Speed (Equal Time)	$v_{avg} = \frac{v_1+v_2+\cdots+v_n}{n}$
Average Velocity	$v\vec{v}_{avg} = \frac{x\vec{v}_f - x\vec{v}_i}{t_f - t_i}$
Instantaneous Velocity	$v = \frac{dx}{dt}$
Instantaneous Acceleration	$a = \frac{dv}{dt} = \frac{d^2x}{dt^2} = v\frac{dv}{dx}$
Velocity-Time Kinematic Eq.	$v = u + at$
Position-Time Kinematic Eq.	$s = ut + \frac{1}{2}at^2$
Position-Velocity Kinematic Eq.	$v^2 = u^2 + 2as$
Displacement in n^{th} Second	$s_{n^{th}} = u - \frac{a}{2}(2n-1)$
Midpoint Velocity	$v_{mid} = \sqrt{\frac{u^2+v^2}{2}}$
Max Height in Free Fall	$H_{max} = \frac{u^2}{2g}$
Time of Flight (Upward Launch)	$T = \frac{2u}{g}$
Splash Hearing Time	$T = \sqrt{\frac{2d}{g}} + \frac{d}{v_s}$

4. Numerical Problem: A train accelerates from rest at $2\ m/s^2$ for 10 s, runs at constant speed for 30 s, and then decelerates uniformly at $1\ m/s^2$ to stop. Calculate the total distance covered and plot the $v-t$ graph.

5. Advanced Challenge: Two trains 100 m and 150 m long head toward each other on parallel tracks at 54 km/h and 72 km/h. Calculate the time taken for them to completely cross each other.